

Bernice E. Rogowitz

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Bernice E. Rogowitz

Ph.D., Columbia University

B. S. Brandeis University

- Psychophysics and Perception

- Visualization, Physicalization, Visual Analysis

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[curriculum vitae](#)

[google scholar profile](#)

Editor in Chief, Journal of Perceptual Imaging

Visual Perspectives Research

(2009-present)

Research in human perception and cognition, in the context of visual analysis, visualization, color imaging, and augmented display technologies, for science, medicine, finance and art. Current projects include (1) a new analog method for physicalizing touches and traces on surfaces and objects, (2) memory for visualizations in 2D, 3D, VR and Mixed Reality systems, and (3) Gestalt principles for spatial-temporal feature analysis.

What's happening

November, 2023 (Paper on Memory in Mixed Reality, Virtual Reality, 3D and 2D)

Our paper, "Memory Recall for Data Visualizations in Mixed Reality, Virtual Reality, 3D and 2D has been accepted by IEEE Transactions in Visualization and Computer Graphics! The idea of this work is explore how different modes of presentation affect how well visualizations, and their spatial location, are recalled. Mixed Reality, where the objects to remember are digitally stitched onto a real 3D room enabled best performance. [Here is a pre-print.](#)

July, 2023 (Dagstuhl Seminar on Art, Illusions and Visualization)

Delighted that our seminar proposal has been accepted by Dagstuhl! It's a wild topic: "Art, Visual Illusions, and Data Visualization." We have an initial invitation list, but would love to hear from you if this hits a resonant chord! [Here is the Seminar announcement.](#)

May, 2023 (Poster on Spatial/Temporal Perception at VSS)

Christophe Hurter (U. Toulouse) and I presented a poster at the Vision Science Society (Meeting) called "Detecting Correlated Target Motion in Moving and Static Dot Arrays." This work looks at how the spatial and temporal characteristics of a subset of dots affects reaction time, building on research in Gestalt psychology, selective attention, and motion masking.

April 2023 (CHI paper published on Touch Perception)

I've been working with Dietmar Offenhuber and Laura Perovich at Northeastern on data physicalization. Our paper, "The Tactile Dimension: a method for physicalizing touch behavior" received an Honorable Mention Award at **CHI 2023**. We describe a new, analog method for capturing and analyzing touch behavior, and provide three user studies validating its reliability and applicability to problems in data physicalization.

January 29, 2023 -February 3, 2023 (Datstuhl Seminar on Perception in Network Visualization)

So excited about the upcoming Dagstuhl seminar on [Perception in Network Visualization](#), Karsten Klein, Steven Kubourov, Danielle Safrir and I are organizing. Almost 40 scientists from diverse disciplines will gather for a week, which we hope will help us advance the research agenda in this emerging topic.

January 15-19, 2023 (Paper presentation t HVEI on Visualization for Public Consumption)

I will be in San Francisco for the Conference on Human Vision and Electronic Imaging (HVEI), which was the launchpad for the [Journal of Perceptual Imaging](#). Sunday, Jan 15, I present a short course on [Perception and Cognition for Imaging](#), and will present a talk on the work Paul Borrel and I have been doing on how people interpret time series COVID data.

February 7, 2022 (JPI update: New Associated Editor)

The [Journal of Perceptual Imaging](#) welcomes Dr. Adar Pelah, York University, to our editorial team. Adar's strong background in perception and proprioception, combined with a keen interest in immersive environment technologies and machine-learning approaches will support the publication of leading edge research in these multidisciplinary areas.

December 8, 2021

Masters' Thesis Committee. I was honored to collaborate with Xinzhe Du and her advisor, Thrasyvoulos Pappas, at Northwestern University. Xinzhe successfully defended her Masters' Thesis today: "Texture Metamers and Perceptual Learning." Congratulations.

October 24, 2021

Paper at Vis2021. Our paper called "Touching Art: A Method for Visualizing Tactile Experience," was presented at the [Alt-Vis Workshop](#) on Sunday, October 24th at 3:00pm Central Time. Here's the link to the preprint: <https://arxiv.org/pdf/2110.00686.pdf>.

October 2, 2021

VisLies. Tune in to VisLies at IEEE Visualization2021, where Ken Moreland and I will be leading another animated session filled with laughter and tears. Check out the VisLies Web Site and Gallery, <http://vislies.org>, and [click here](#) to sign up to present.

Tuesday, October 26 at 3:00 CDT.

For previous happenings, [click here](#)

[Research Areas in Visualization and UI Design](#)-- why perception matters



Journal of Perceptual Imaging

Journal of Perceptual Imaging (JPI)

Co-Founder and Editor-in-Chief

(2019-present)

The [Journal of Perceptual Imaging](#) publishes multidisciplinary research that explores how principles of perception and cognition support and inspire new technologies, and how emerging technologies drive new questions for perceptual research. Experimental, theoretical, algorithmic, and survey papers are welcome. JPI is an open access, peer-reviewed publication of the Imaging Science and Technology Society (IS&T). [Submit to JPI](#)



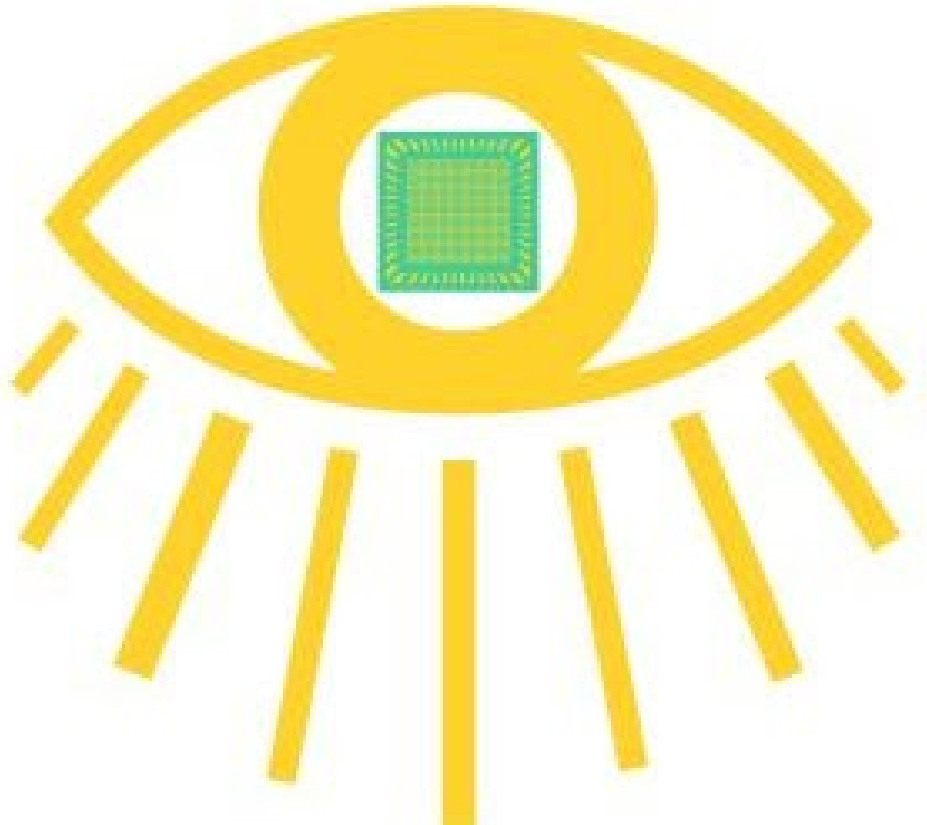
Columbia University

Lecturer, School of Professional Studies

Masters Program in Applied Analytics

(2017-2019)

I created the curriculum for, and lectured in, the core course in Data Visualization and Design, which taught visualization for communication and for analysis through hands-on projects with real data.



Conference on Human Vision and Electronic Imaging (HVEI)

Founder and Co-Chair

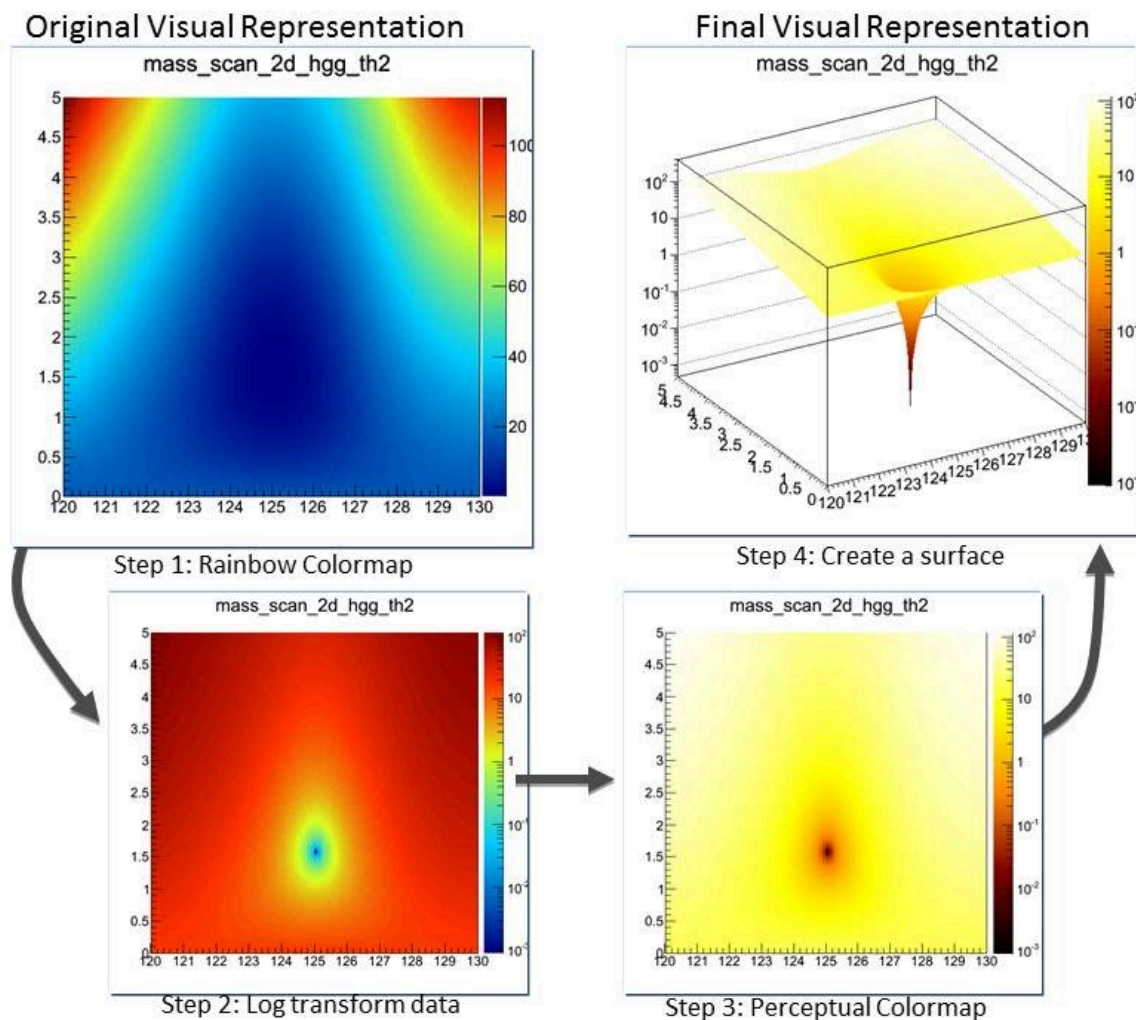
(1989-2018)

I founded HVEI in order to build a multidisciplinary community of scientists and technologists working at the intersection between perception, cognition and emerging electronic technologies. The topics evolved over the years, reflecting the evolution of technology. The reach continues to be broad under the current leadership, including early vision models, digital coding, image quality, attention, visualization, multisensory interfaces, brain imaging, art and design. [HVEI 25th Anniversary Talk:](#)

[Finding Features in Data - Bernice Rogowitz](#)

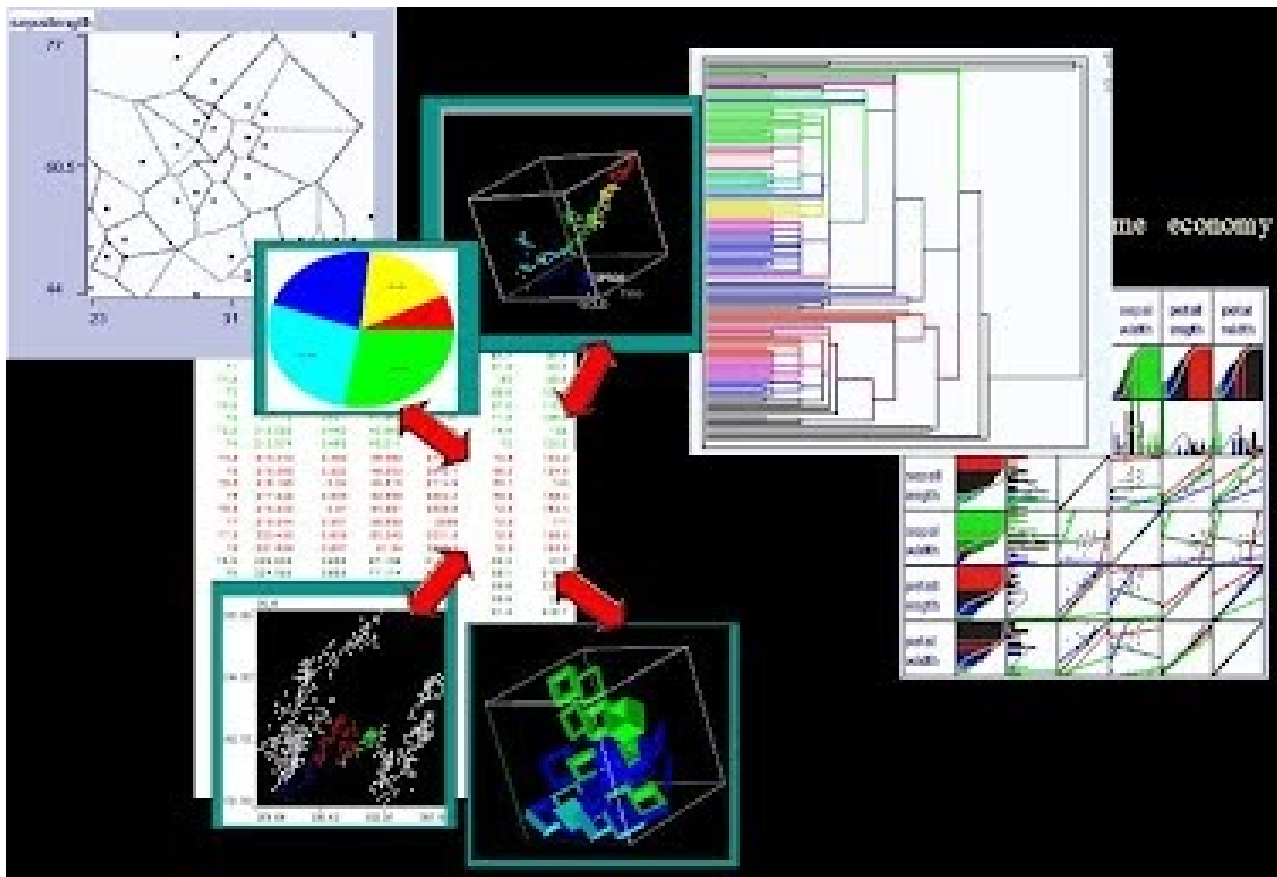
The same data with two different color maps. The rainbow colormap (left) obfuscates the features in this visualization of Florida; the perceptual colormap (right) provides a more faithful representation of the data.

[Rogowitz and Treinsh, 1998.](#)



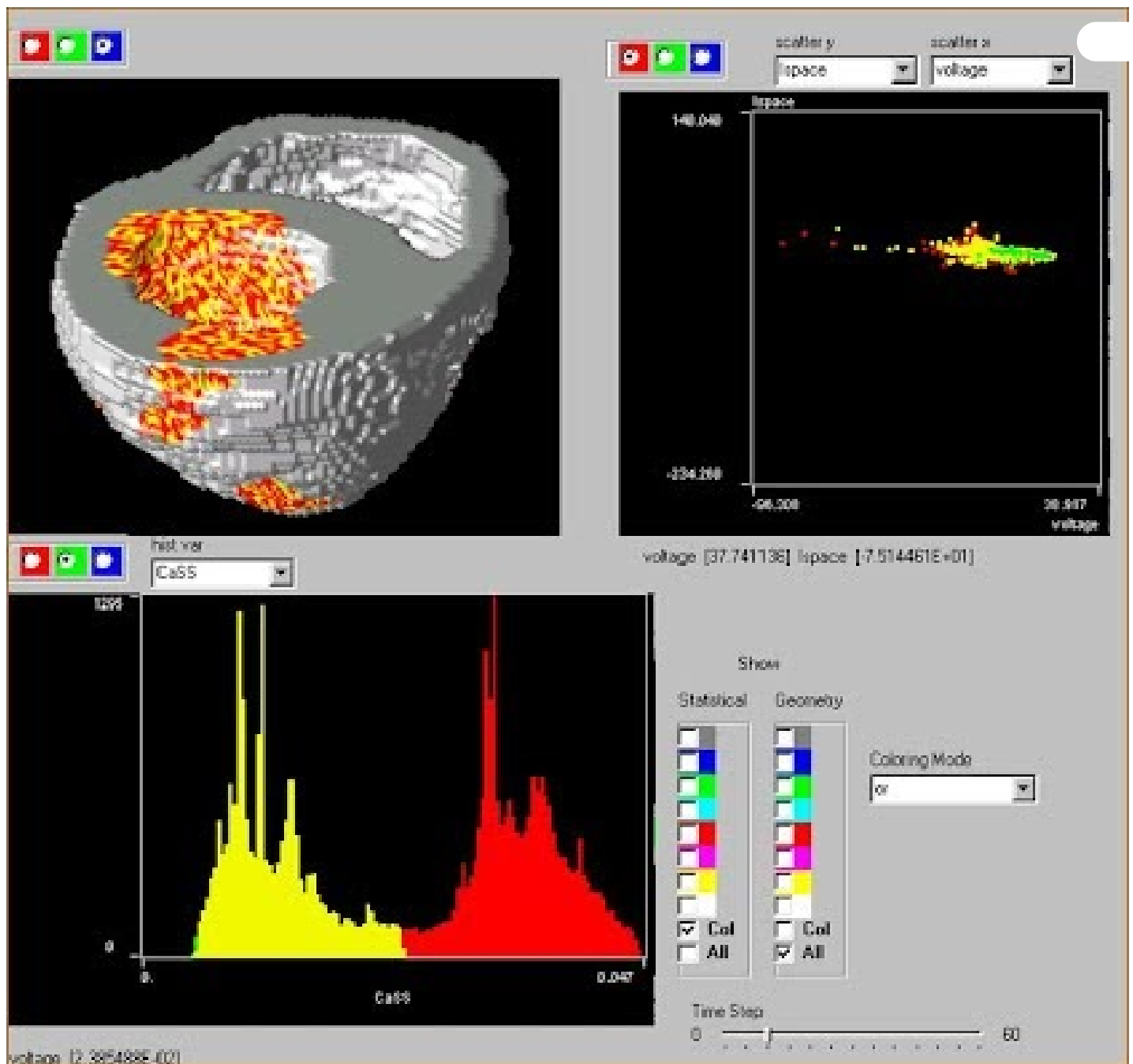
In the Higgs-Boson experiment, two specially-constructed beams of particles were set to collide in front of a particle detector. All four panels show the mass (GeV/c²) of hundreds of trillions of collisions. The hypothesized Higgs-Boson was predicted to have a mass of 125-127 Giga Electron Volts. The top left panel uses a rainbow colormap to represent the mass across the detector. The data were log transformed (bottom left), then a colormap with monotonic luminance was used (bottom right), and, in the top right, the mass was redundantly mapped onto a height, producing a visualization that clearly shows the mass signature of the Higgs Boson. [Rogowitz and Couet, CERN 2013](#)

[Interactive Visualization](#) - the Visualization and Visual Analysis (ViVA) Workbench



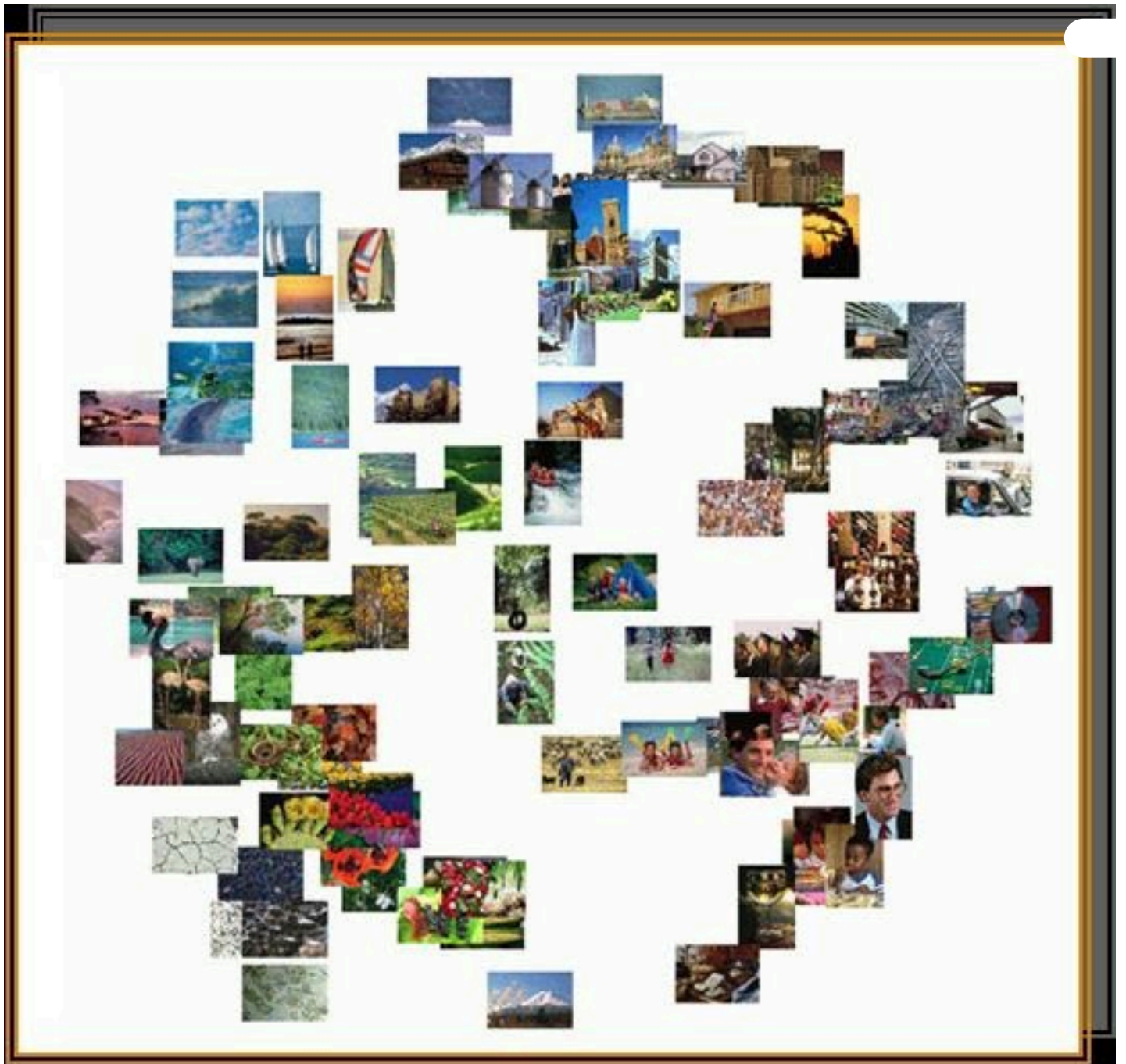
At IBM Research, [my.group](#) developed an interactive visualization and analysis package, which we licensed to SPSS and used in IBM offerings, such as [FAMS](#). ViVA offers over 20 linked visualizations, with color brushing, plus a library of over 250 mathematical functions. [Download it free from Source Forge!](#)

Finding Patterns across Multiple Data Types



Working with Johns Hopkins University, we developed a method for linking a 3-D model of the heart with 2-D visualizations of parameters of an underlying finite element model. A 3-d model of a heart is shown in the top left panel. For every point in the heart, there are values of a finite element model, which are visualized in the scatterplot and histogram. In this example, the analyst has marked two regions in the histogram, corresponding to two different ranges of the variable 'Calcium in the Subspace of the Heart Organelle' and can immediately see which areas of the 3-D heart have these values. [Gresh, Rogowitz, Winslow, Scollan, Yung, 2001.](#)

[Image Semantics and Search](#)



[Video Micro-Tags](#)

[Surveillance](#)

RECENT PUBLICATIONS

Rogowitz, B.E., Topkara, M., Pfeifer, W., and Hampapur, A., [Perceptual Evaluation of Visual Alerts in Surveillance Videos](#), Proceedings of the SPIE, Conference on Human Vision and Electronic Imaging (2015)

Rogowitz, B. E., "[Perceptual approaches to finding features in data](#)," Proceedings of the SPIE, Vol 8651, Conference on Human Vision and Electronic Imaging XVIII (2013)

Rogowitz, B.E. and Goodman, A., "[Integrating human- and computer-based approaches for feature extraction and analysis](#)," Proceedings of the SPIE, Vol. 8291, SPIE/IS&T Conference on Human Vision and Electronic Imaging XVII (2012), 82910W.

Rogowitz, B.E. and Matasci, N., ""[Metadata mapper: a user interfaces web service for mapping data between independent visual analysis components, guided by perceptual rules](#)," Proc. of the SPIE, Vol. 7865 (2011)

Rogowitz, B.E., Pappas, T, and Allebach, J. "Human Vision and Electronic Imaging," chapter in Handbook of Optics, edited by Bass, M., Enoch, J, and Lakshminarayanan, V. Volume III: Vision and Optics, McGraw Hill, 2010

RECENT HONORS AND AWARDS

2015 IS&T Honorary Member

Honorary membership, the highest award bestowed by the Society, recognizes outstanding contributions to the advancement of imaging science or engineering.

Bernice E. Rogowitz

for outstanding leadership, teaching, research, and building a multi-disciplinary community of scientists and technologists as conference chair of the Human Vision and Electronic Imaging Conference

Bernice Rogowitz is a multidisciplinary scientist, working at the intersection of human perception, imaging, and visualization. Dr. Rogowitz received her BS in experimental psychology from Brandeis University and PhD in vision science from Columbia University. She was a Post-Doctoral Fellow in the Laboratory for Psychophysics at Harvard University.

For many years, she was a scientist and research manager at the IBM T.J. Watson Research Center, and is currently active in research and teaching through her consulting company, Visual Perspectives. Her work includes fundamental research in human color and pattern perception, novel perceptual approaches for visual data analysis and image semantics, and human-centric methods to enhance visual problem solving in medical, financial, and scientific applications. As the founder and co-chair of the IS&T conference on Human Vision and Electronic Imaging, she has been a leader in defining the research agenda for human-computer interaction in imaging, driving technology innovation through research in human perception, cognition, and aesthetics.

SPIE Fellow, January 20, 2009:

"Dr. Rogowitz's fellow promotion is for specific achievements in human vision applications in electronic imaging. She has been a leader in research in perceptual areas relevant to imaging and visualization systems, in developing interactive software systems based on her deep understanding of these issues, and building a community linking the engineering and human vision communities.

Her research in human and machine vision and image processing has included important contributions to many aspects of spatial vision, spatial-temporal interactions, shape perception, color vision, and image perception. Rogowitz has applied this fundamental knowledge to a wide variety of engineering tools and applications from measuring display artifacts, to representing meaning in data visualization, to navigating more naturally through a digital archive of images, to conveying the sense of touch in 3-D virtual environments.

Dr. Rogowitz is very active in the optics community both with her collaborative work and with scientific societies. She is a fellow of the IS&T, a senior member of IEEE, and an active leader with SPIE. She is the founder and co-chair of the SPIE/IS&T Conference on Human Vision and Electronic Imaging. For her work she has been honored as a National IS&T Distinguished Lecturer and received the IS&T/SPIE Award in Recognition of Outstanding Achievement."

[Bernice @ Glance](#)

[IS&T Conference on Human Vision and Electronic Imaging](#) (1989-present)





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